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NPB 100L A02

Effect of Cooling on Muscle Response due to Achilles Tendon Reflex

Abstract:

The Achilles tendon reflex is a monosynaptic stretch reflex commonly used to assess peripheral nerve function and neuromuscular integrity. This study investigated the effect of localized cooling on reflex latency in thirteen healthy college students. Surface electromyography (EMG) was recorded from the gastrocnemius muscle during baseline (room temperature) and after 10 minutes of ice-pack cooling applied to the posterior calf. Reflex latency was quantified as the time from tendon tap to EMG onset. Cooling resulted in a consistent prolongation of latency across subjects, with mean latency increasing from 36.3 ms at baseline to 38.0 ms post-cooling (~1.7 ms increase). These changes likely reflect both slowed axonal conduction due to reduced ion channel kinetics and temperature-dependent slowing of synaptic and neuromuscular transmission. Our results support the conclusion that local cooling significantly modulates multiple components of the stretch reflex pathway and highlight the sensitivity of reflex timing to thermal perturbation.

Introduction:

The Achilles tendon reflex is a classic monosynaptic stretch reflex used to assess peripheral nerve function, synaptic transmission, and neuromuscular integrity. When the Achilles tendon is tapped, muscle spindle afferents in the calf are rapidly stretched, generating an action potential that propagates proximally along afferent fibers to the spinal cord. Following a brief synaptic delay, α-motor neurons activate the gastrocnemius and soleus muscles, producing a measurable electromyographic (EMG) response. The latency of this response reflects two distinct temporal components: (1) a distance-dependent axonal conduction time determined by conduction velocity along peripheral nerves, and (2) a distance-independent fixed delay (t_{fixed}) that includes synaptic transmission, neuromuscular junction delay, and electromechanical coupling before force generation. Thus, total EMG latency can be expressed as:
Latency = $t_{fixed} + d * v_{axon}$ where d is the conduction distance and v_{axon} is the axonal conduction velocity.

Temperature is a critical physiological modifier of nerve and muscle function. Cooling peripheral nerves slows the kinetics of voltage-gated ion channels, reducing the rate of depolarization and thereby decreasing axonal conduction velocity. As a result, cooled nerves typically exhibit prolonged response latencies. However, cooling may also influence the fixed-delay components of the reflex arc, particularly synaptic transmission speed, neuromuscular junction efficiency, and excitation–contraction coupling. This can potentially amplify the total increase in latency beyond what would be expected from slowed conduction alone.

Experimental work examining the effects of local cooling on neuromuscular performance supports these possibilities. Halder et al. (2015) demonstrated that 20 minutes of cooling the lower leg in 10°C water significantly reduced maximal isometric strength in the tibialis anterior and altered EMG characteristics in the gastrocnemius medialis. Specifically, the gastrocnemius exhibited increased EMG amplitude following cooling, which the authors interpreted as evidence of neuromuscular fatigue or altered muscle activation strategies. Although gait ground reaction forces remained unchanged during submaximal locomotion, the findings show that local cold exposure can induce measurable neuromuscular changes even in healthy individuals. These results support the idea that cooling affects both peripheral nerve conduction properties (e.g., reduced ion channel kinetics) and muscle activation behavior, both of which may contribute to changes in reflex latency..

Together, prior work suggests that cooling may have significant and multifaceted effects on the latency of the Achilles tendon stretch reflex. Reduced temperature is known to slow the kinetics of voltage-gated ion channels, thereby decreasing peripheral axonal conduction velocity and increasing the time required for sensory afferents to propagate the stretch signal to the spinal cord. Evaluating how cooling influences latency provides insight into the broader physiological impact of temperature on neural and neuromuscular function. Based on prior evidence that cooling slows axonal conduction and affects neuromuscular transmission, we hypothesized that local cooling of the posterior calf would significantly increase Achilles tendon reflex latency compared to baseline room-temperature conditions.

Methods:

Participants

Thirteen healthy college student volunteers participated in this study. All procedures were completed within approximately 20 minutes per subject.

Instrumentation and Electrode Preparation

Surface EMG activity was recorded using non-adhesive electrodes placed over the thickest portion of the gastrocnemius muscle in the posterior calf. After securing the electrodes, the surrounding skin was marked to ensure consistent placement between baseline and post-cooling measurements. A handheld ground electrode was provided to each subject to minimize electrical noise during signal acquisition. To standardize conduction distance across subjects, the length from the electrode site to a reference point on the spine (four finger-widths above the end of the tailbone) was measured. This reproducible measurement served as an estimate of the nerve pathway length relevant to reflex latency.

Baseline Procedure

Participants were seated with their feet hanging freely off the edge of a table to maintain a relaxed Achilles tendon and to provide unobstructed access for mechanical stimulation. Three

Baseline EMG recordings were collected under room-temperature conditions. Achilles tendon reflexes were elicited using a standard reflex hammer tapped gently but consistently at the tendon to evoke a muscle response. The baseline measurements served as each subject's reference for conduction velocity and fixed-delay components of the reflex.

Cooling Procedure

After baseline trials, subjects cooled the posterior calf around the electrode region using an ice pack for 10 minutes. Following cooling, the skin was thoroughly dried to ensure proper electrode function and electrode placement was verified using the markings established earlier. Subjects remained seated with feet freely hanging for consistency with the baseline condition.

Post-Cooling Reflex Recording

The same stimulation protocol used during baseline was repeated after cooling. Three additional Achilles tendon reflex trials were collected per subject. Effort was made to maintain identical stimulation parameters, including hammer strike intensity, timing, and trial spacing, to ensure that temperature was the primary variable affecting reflex latency.

Data Collection and Latency Measurement

EMG signals were displayed on an oscilloscope, and reflex latency was quantified as the time between the onset of the mechanical tap and the onset of the muscle response. These latency measurements from both conditions were used for subsequent analysis of conduction velocity and fixed-delay components.

Data Analysis

For each participant, the mean latency across the three trials per condition (baseline and post-cooling) was computed to reduce intra-subject variability. A paired t-test was used to compare reflex latency before and after cooling within subjects. This test evaluated whether local cooling caused a statistically significant change in mean latency. Statistical significance was assessed at $p = 0.05$.

Individual and mean reflex latencies were visualized using line plots (paired baseline vs. post-cooling) and bar plots with error bars representing ± SEM, providing a clear depiction of both within-subject changes and overall trends. There were no overlapping ranges for the SEM indicating that our data was statistically significant.

Results:

Reflex latency increased following local cooling of the posterior calf. Figure 1 shows individual paired latency measurements ($n = 13$) collected at room temperature and after 10 minutes of cooling, along with the group mean ± SEM. At room temperature, mean Achilles tendon reflex latency was 36.35 ± 0.77 ms, whereas post-cooling latency increased to 37.95 ± 0.70 ms,

reflecting an average prolongation of ~1.6 ms. All participants demonstrated an upward shift in latency, indicating a consistent slowing of the reflex response after cooling.

A paired t-test confirmed that this increase was statistically significant. Because our hypothesis predicted a unidirectional effect, such that cooling would increase latency, we evaluated the data using a one-tailed paired t-test, which yielded $p = 2.35 \times 10^{-7}$. This extremely small p-value demonstrates overwhelming evidence that reflex latency increased following cooling, far beyond what would be expected from random within-subject variability. Furthermore, the paired t-test yielded a t-statistic of |9.75|, indicating that the cooling-induced increase in reflex latency was nearly ten standard errors away from zero and therefore represents an exceptionally strong and consistent effect across participants.

The minor inter-subject variability in latency values, visible in the spread of individual traces was expected given natural anatomical differences in conduction distance and neuromuscular physiology. However, the consistent direction of change across subjects strengthens the interpretation that cooling exerts a measurable physiologic effect on the Achilles tendon stretch reflex.

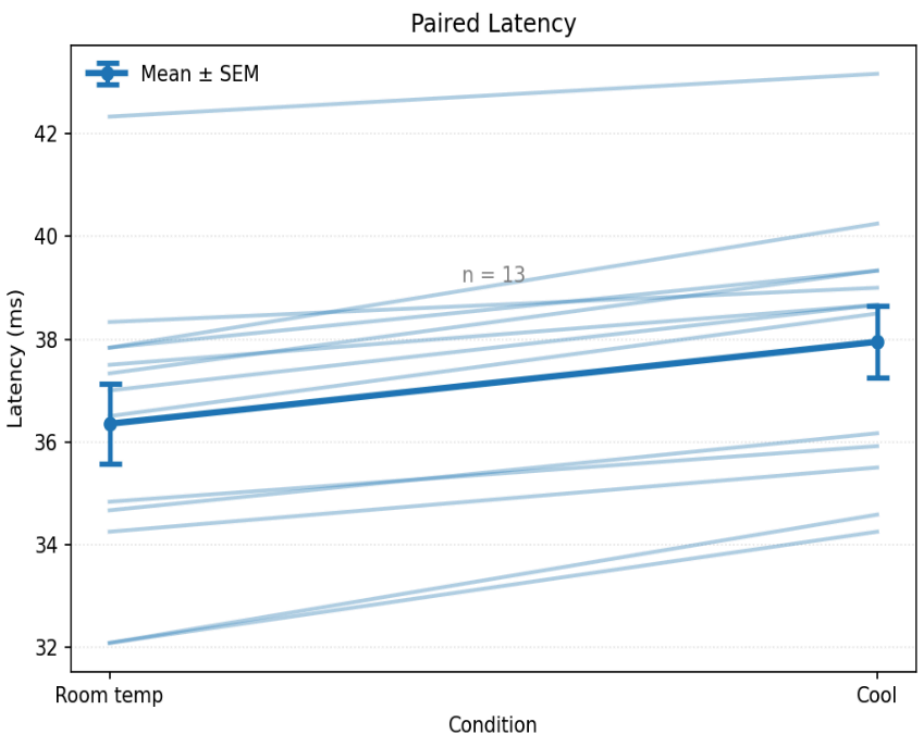

Figure 1. Paired Achilles tendon reflex latencies at room temperature and after cooling. Individual subject latencies (thin light-blue lines) and group mean ± SEM (thick dark-blue line with error bars) are shown for the room-temperature and cooled conditions (n = 13). Reflex latency increased for nearly all subjects following 10 minutes of local calf cooling, with the

group mean increasing from 36.35 ± 0.77 ms at baseline to 37.95 ± 0.70 ms after cooling. This pattern indicates a temperature-dependent slowing of the stretch reflex, consistent with reduced axonal conduction velocity and potential changes in synaptic and neuromuscular transmission.

Discussion:

Our findings that cooling increases overall Achilles reflex latency align with broader evidence that temperature modulates neural conduction and reflex timing. Total latency reflects both axonal conduction velocity and fixed-delay components including synaptic transmission, neuromuscular junction delay, and excitation–contraction coupling so the observed prolongation is consistent with the known slowing of peripheral nerve function under reduced temperature (Racinais & Cresswell, 2013).

Cooling slows the kinetics of voltage-gated ion channels, thereby decreasing conduction velocity along afferents and α-motor neurons, which contributes to the steeper latency–distance slope observed in cold conditions. Beyond pure axonal effects, localized cooling has also been shown to transiently reduce motor nerve conduction velocity and alter motor unit firing behavior, demonstrating that regional temperature reductions can affect peripheral neural transmission independent of voluntary effort (Richards et al., 2021). Classical studies of the H-reflex and tendon-tap reflex further indicate that cooling alters reflex waveform properties and can reduce muscle spindle–mediated activity (Bell & Lehmann, 1987), suggesting that synaptic and afferent transmission stages of reflex pathways are also temperature-sensitive (Dewhurst et al., 2005).

Taken together, these findings support the conclusion that cooling alters multiple components of the Achilles reflex latency. Slower axonal propagation due to cooler nerve environments, combined with possible contributions from altered synaptic and neuromuscular transmission, increases total reflex latency. This mechanistic understanding reinforces the utility of dissecting reflex latency into conduction and fixed-delay components to better interpret neuromuscular responses to thermal perturbations.

Limitations:

Despite the clarity of the observed effects, several limitations constrain the interpretation of our findings. The sample size of thirteen participants, although appropriate for a within-subject design, limits the generalizability of the results to larger or more diverse populations. Thus, we could not mathematically separate changes in axonal conduction velocity from alterations in fixed-delay components such as synaptic transmission or neuromuscular junction delay. Moreover, the study utilized only a single cooling duration and intensity, restricting our ability to evaluate how different temperatures or cooling protocols might differentially influence reflex latency. The use of surface EMG, while practical and reliable for onset detection, prevents a detailed analysis of deeper muscle activation patterns or motor unit–specific contributions. Although efforts were made to maintain consistent tendon tap force, small unavoidable variations

in mechanical stimulation could introduce minor variability in latency measurements. These methodological constraints should be considered when interpreting the findings.

Future Applications:
These insights could have meaningful clinical implications, particularly in the assessment of neuropathies, demyelinating disorders, or cryotherapy-based rehabilitation protocols, where the temperature dependence of conduction velocity is diagnostically relevant (Racinais & Cresswell, 2013; Richards et al., 2021). Integrating Achilles tendon–based reflex measures with electrically evoked H-reflex or M-wave responses may also help isolate afferent, synaptic, and efferent components of the latency, an approach supported by earlier work showing that cooling alters both tendon-tap reflexes and electrically evoked reflex pathways (Bell & Lehmann, 1987; Dewhurst et al., 2005). Beyond clinical contexts, this knowledge may inform athletic performance and injury-prevention strategies, as cooling is frequently used in sports medicine and is known to acutely modify neuromuscular responsiveness and muscle activation characteristics (Halder et al., 2015; Racinais & Cresswell, 2013).

Conclusion:

Local cooling of the posterior calf significantly increases Achilles tendon reflex latency, demonstrating the temperature sensitivity of peripheral neural and neuromuscular function. The observed prolongation reflects both reduced axonal conduction velocity and potential slowing of synaptic and neuromuscular transmission, emphasizing that total reflex latency comprises multiple temperature-sensitive components. These findings reinforce the value of separating conduction and fixed-delay contributions when interpreting reflex measurements and suggest that even brief local cooling can measurably alter neuromuscular responses. Understanding these thermal effects has implications for clinical neuromuscular assessment, rehabilitation, and performance under varying environmental conditions.

References

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